Cooking with renewable energy

Cucinare con l’energia pulita

Renewable energy comes from sources that do not run out: the sun, the wind, flowing water and plant or food waste. This guide explains how each one can reach your kitchen, and the cooking habits that use less energy whatever your fuel.

1. Solar cooking

A solar cooker uses shiny reflectors to focus sunlight onto a dark pot inside a clear cover. The dark pot absorbs the light and the cover traps the heat. On a sunny day it can slow-cook beans, stews and rice with no fuel.

It works best in the middle of the day with strong sun, for dishes that like long, gentle cooking.

2. Green electricity

Electric and induction hobs can run on electricity generated by wind turbines, solar panels and hydroelectric dams. You can choose a renewable electricity tariff, or produce your own with solar panels.

Induction is the most efficient way to cook with electricity, because it heats the pan itself.

3. Biogas from food waste

When food scraps are collected separately, they can be broken down in sealed tanks by bacteria (anaerobic digestion). This makes biogas, which can generate electricity or heat, and a fertiliser for farms.

Putting scraps in the food-waste bin, not general rubbish, helps this happen.

4. Heat-retention cooking

Bring a pot to the boil, then put it in an insulated box or wrap it well in blankets. The food keeps cooking with its own heat. Italian families once used a "cassetta di fieno", a box packed with hay.

It suits beans, soups, grains and stews. Keep food hot (above 60°C) and reheat it to boiling before eating.

5. Habits that save energy

- Put a lid on the pot.
- Match the pan size to the ring.
- Boil only the water you need.
- Turn off the heat a few minutes early and let the leftover heat finish the job.
- Batch cook and freeze portions.

- Fill the oven when it is on.
- Cook seasonal, local food, which needs less energy to grow, store and transport.

Find out more

Sources and further reading are listed on the FoodFusion Educational Resources page.